

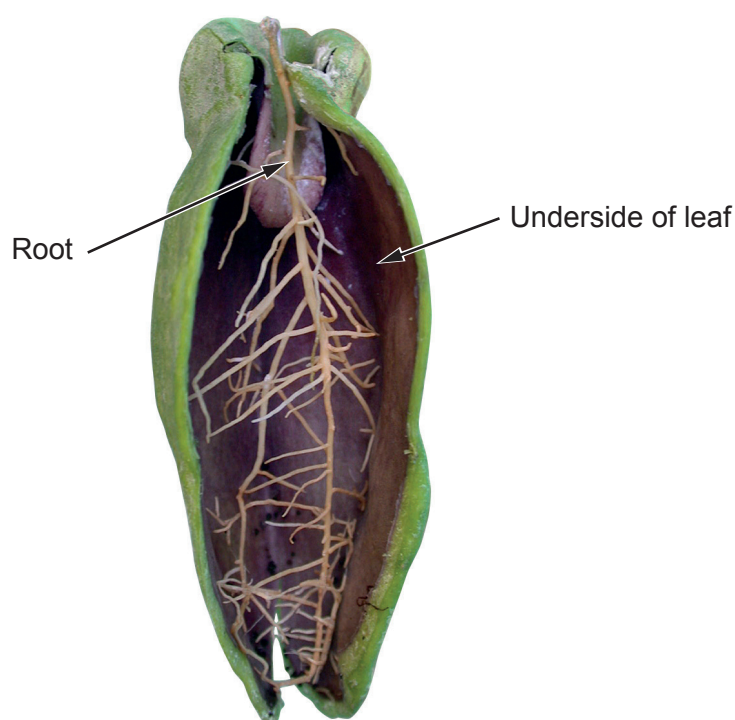
3. *Dischidia major* is a plant found in the rainforests of Malaysia. It is an epiphyte, a non-parasitic plant that grows on other plants.

Some of the leaves of *Dischidia* roll up to form pod-like structures (**photograph 1**).



Photograph 1

The roots of *Dischidia* grow into the pods. **Photograph 2** shows an opened pod to show the underside of the leaf and the roots.



Photograph 2



Examiner
only

(a) Ants of the genus *Philidris* live inside these pods. The ants store dead insects and rear their young inside the pods. The ants and dead insects contribute to the nutrients available to these leaves.

(i) The main source of carbon for these leaves is from the air inside the pod, not the atmosphere. Describe how carbon inside the pod is made available for use by these leaves. [4]

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(ii) Describe how nitrogen is made available for use by these leaves. [3]

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(b) In areas of Malaysia large-scale deforestation is taking place.

(i) Explain how this deforestation could **increase** the concentration of carbon dioxide in the atmosphere in the area. [2]

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(ii) Explain how this deforestation could **decrease** the concentration of carbon dioxide in the atmosphere in the area. [1]

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